Component Interaction Diagrams

Detailed Mermaid diagrams documenting recmeet's architecture, component interactions, and internal control flow. These diagrams reflect the actual source code implementation, not aspirational design.

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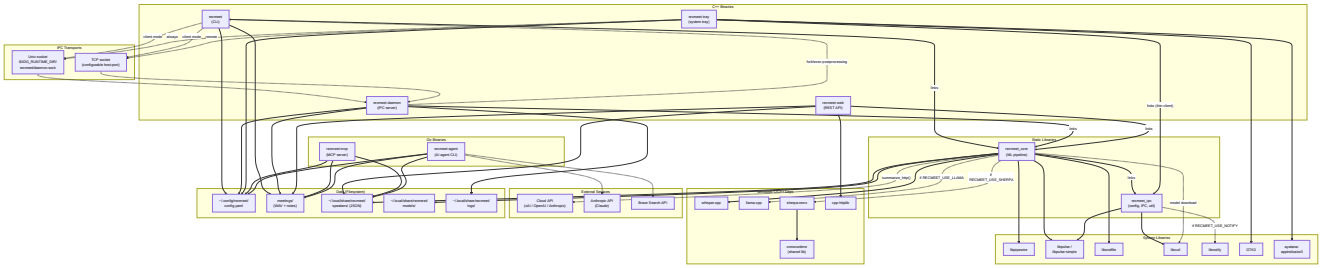
1. Top-Level Component Interaction

All runtime communication paths between binaries, libraries, and external systems. Solid lines are compile-time links; dashed lines are runtime IPC or network calls.



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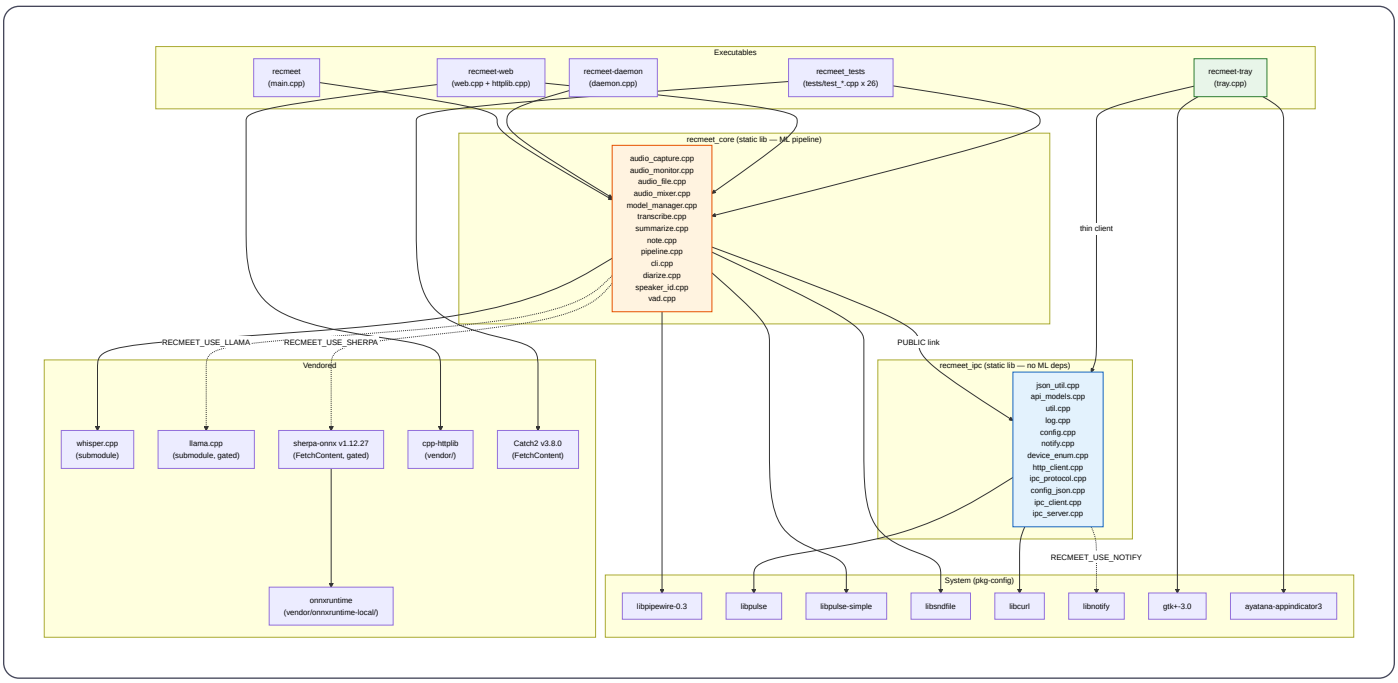
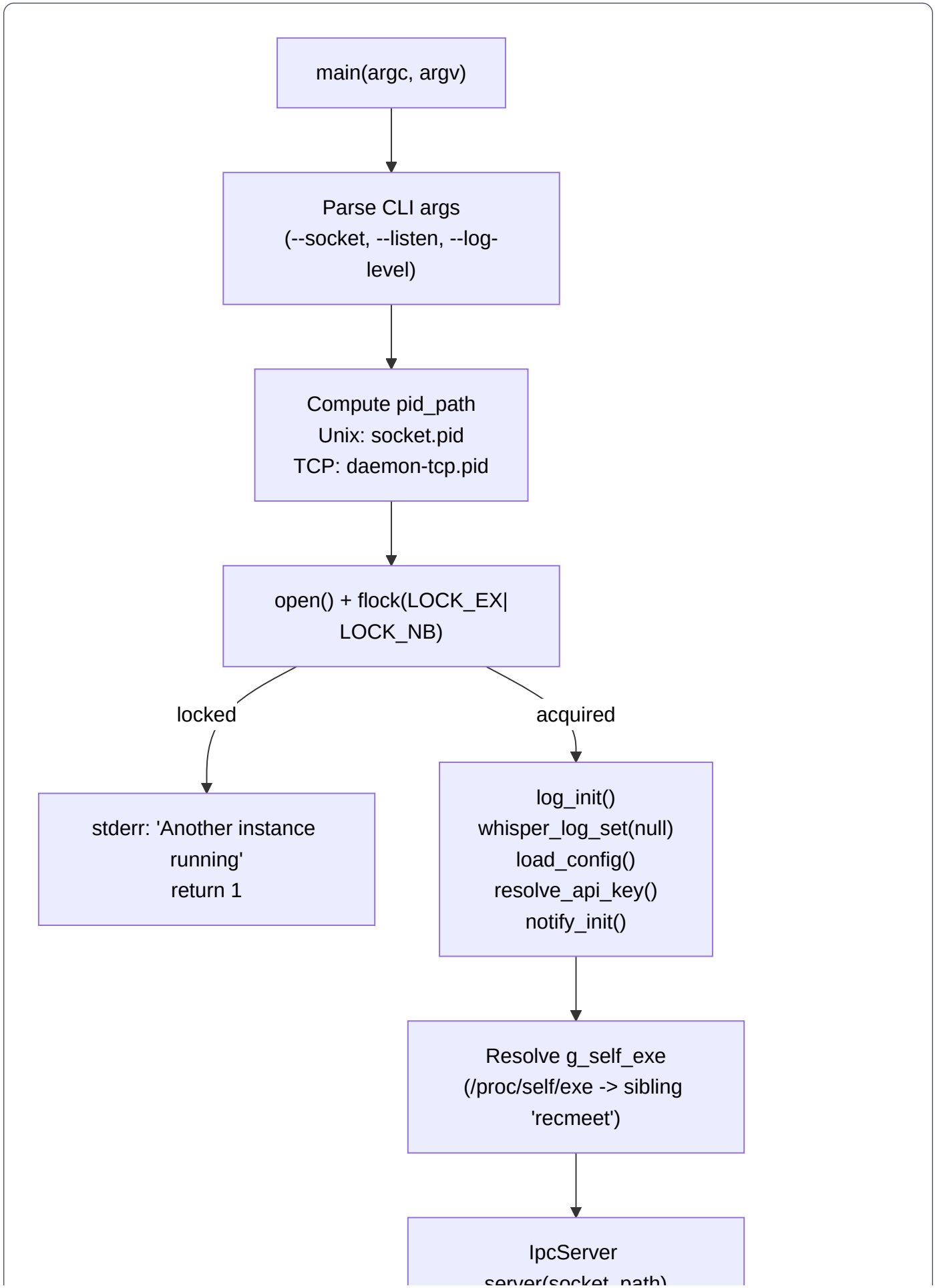
2. Build Topology and Library Dependencies

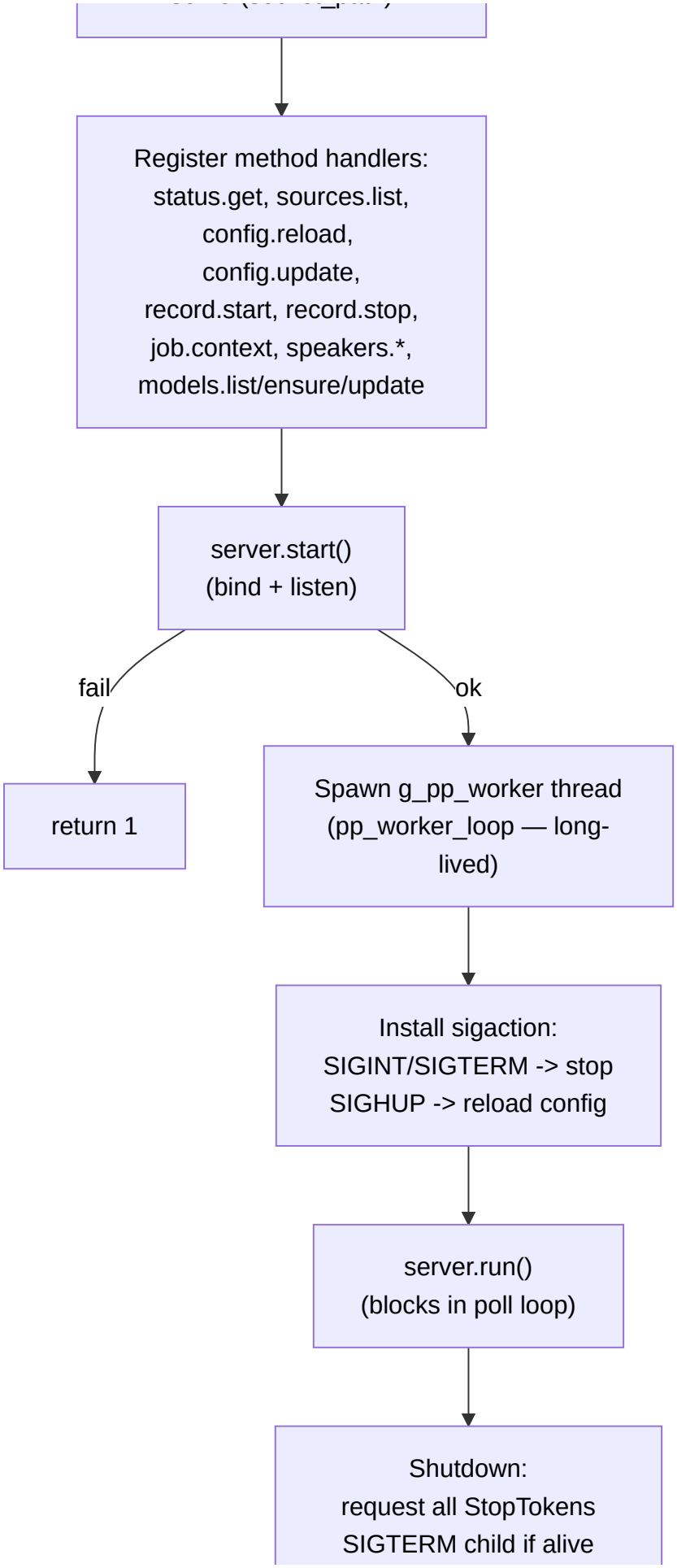
Exact source file inventory per CMake target and the link dependency graph.

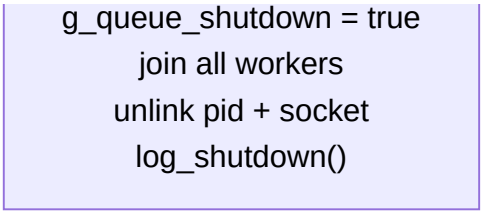


3. Daemon Internals

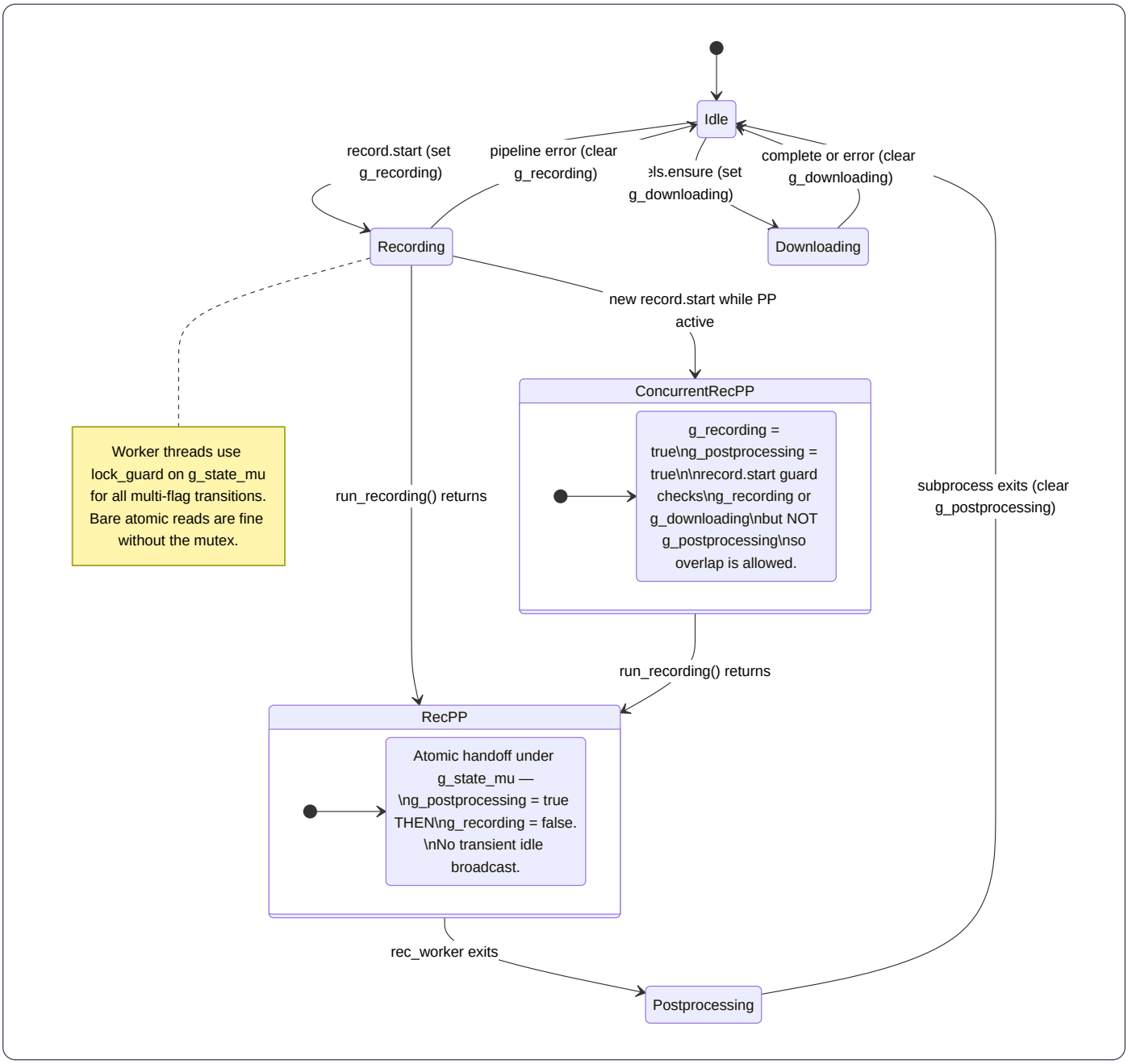
3a. Daemon Startup Sequence


main(argc, argv)
Parse CLI args (--socket, --listen, --log-level)
Compute pid_path Unix: socket.pid TCP: daemon-tcp.pid
open() + flock(LOCK_EX|LOCK_NB)
locked
acquired
stderr: 'Another instance running' return 1
log_init() whisper_log_set(null) load_config() resolve_api_key() notify_init()
Resolve g_self_exe (/proc/self/exe -> sibling 'recmeet')
IpcServer server(socket_path)

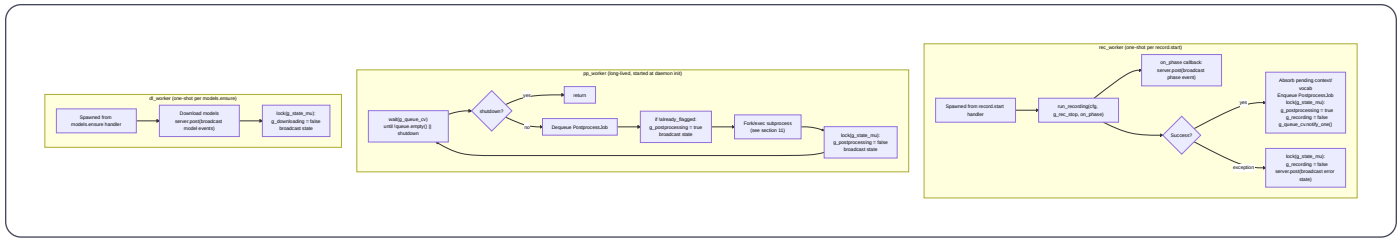

Register method handlers:
status.get, sources.list,
config.reload,
config.update,
record.start, record.stop,
job.context, speakers.*,
models.list/ensure/update
server.start()
(bind + listen)
fail
ok
return 1
Spawn g_pp_worker thread
(pp_worker_loop — long-lived)
Install sigaction:
SIGINT/SIGTERM -> stop
SIGHUP -> reload config
server.run()
(blocks in poll loop)
Shutdown:
request all StopTokens
SIGTERM child if alive



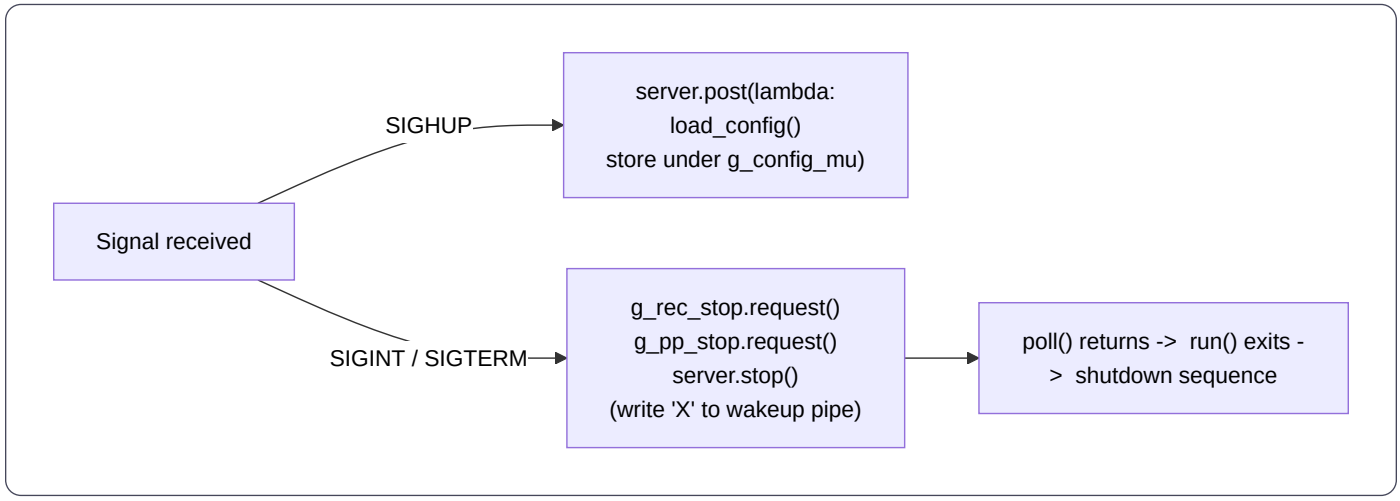
3b. Daemon State Machine



3c. Worker Thread Lifecycle



3d. Signal Handling



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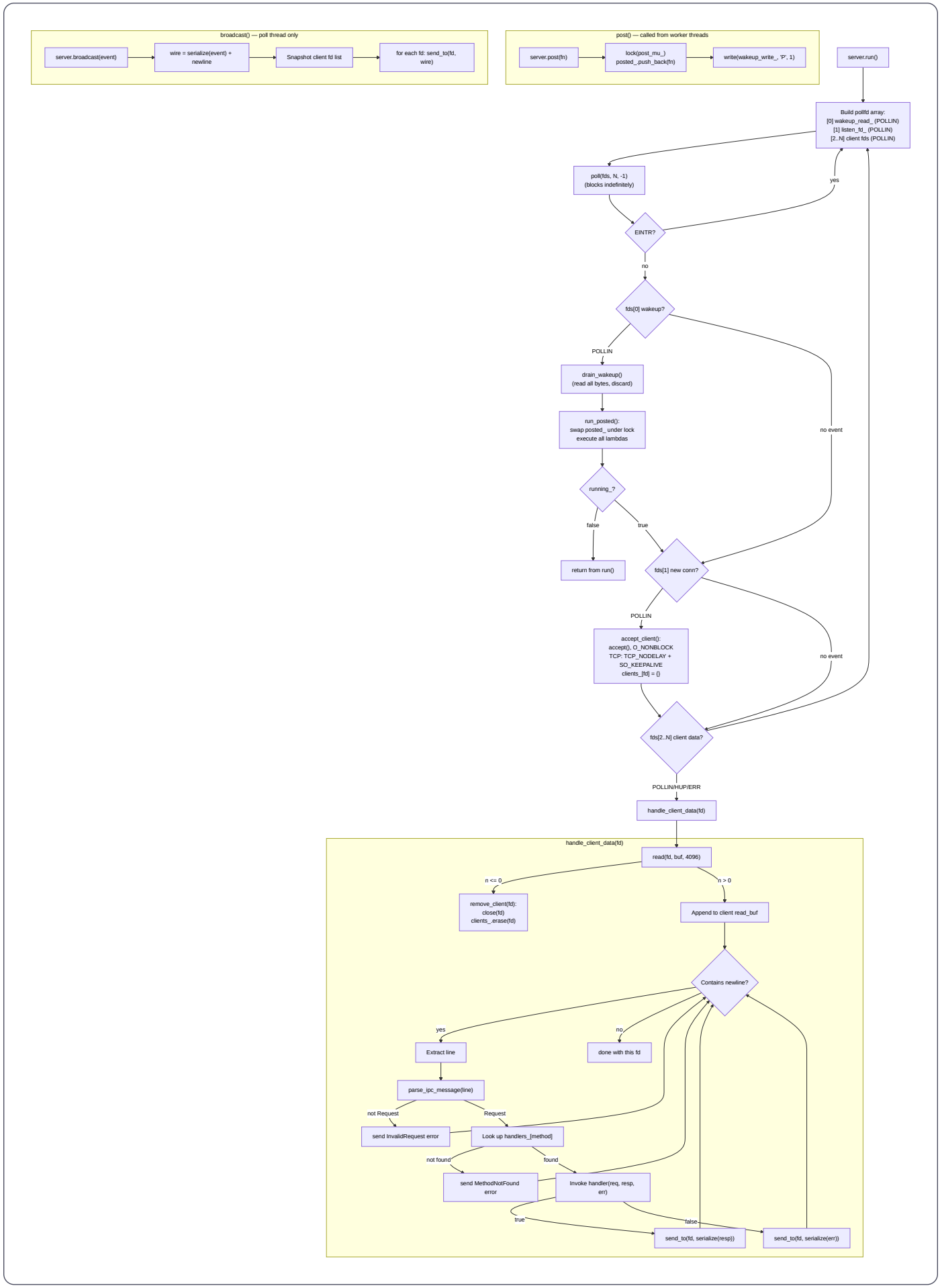
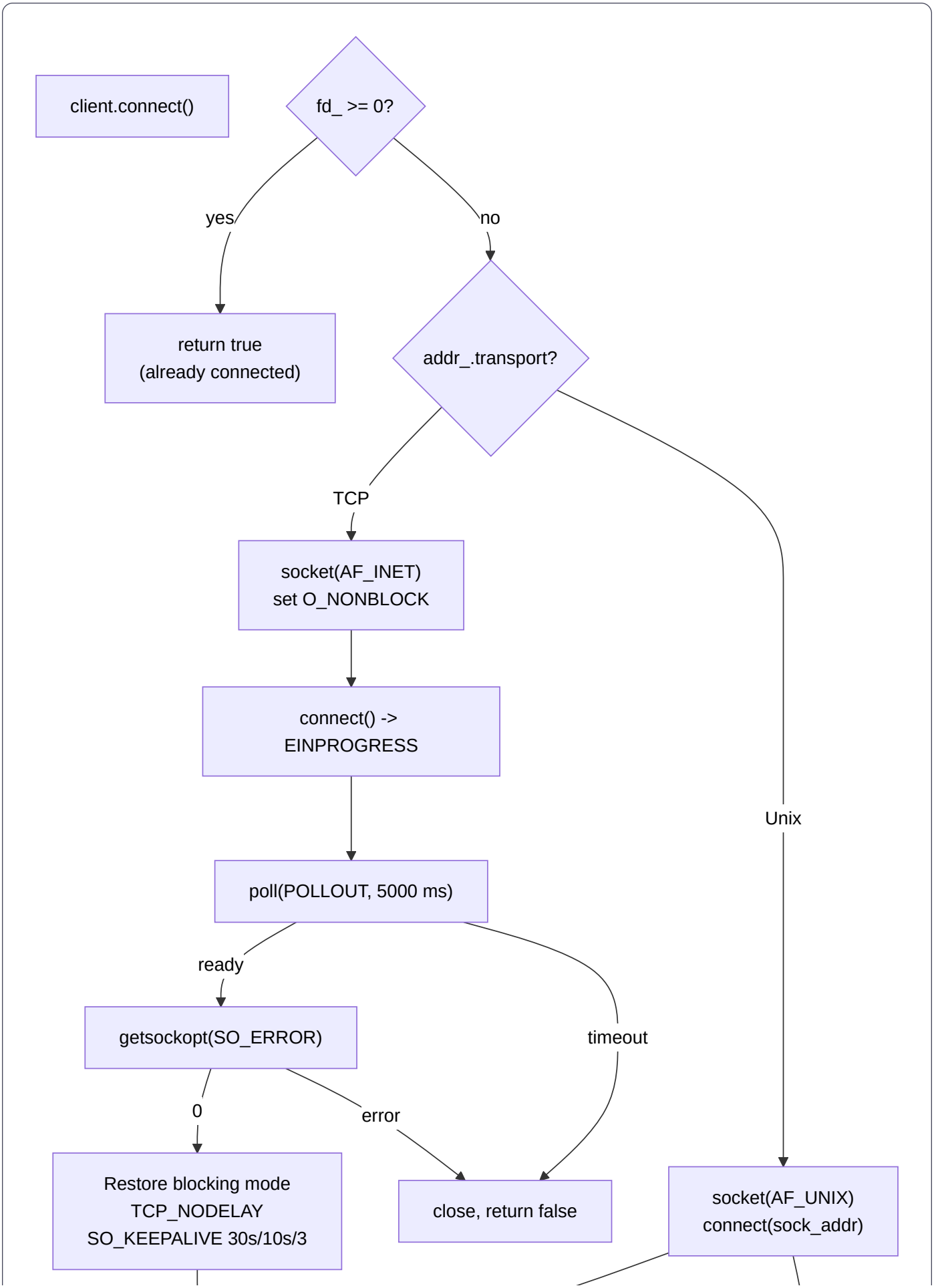
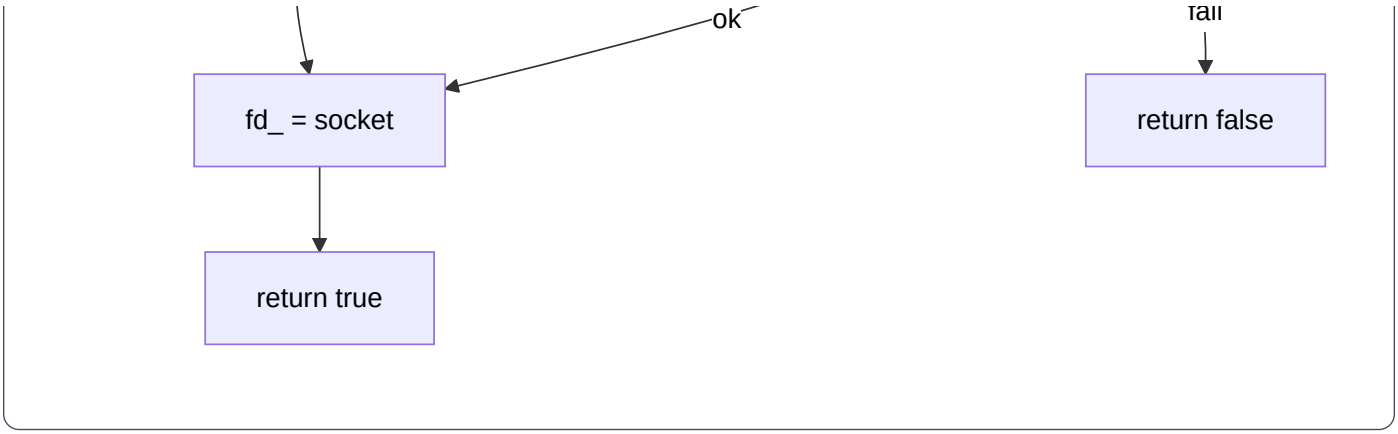
4. IPC Server Poll Loop


broadcast() — poll thread only
server.broadcast(event)
wire = serialize(event) + newline
Snapshot client fd list
for each fd: send_to(fd, wire)
post() — called from worker threads
server.post(fn)
lock(post_mu_) posted_.push_back(fn)
write(wakeup_write_, 'P', 1)
server.run()
Build pollfd array: [0] wakeup_read_ (POLLIN) [1] listen_fd_ (POLLIN) [2..N] client fds (POLLIN)
poll(fds, N, -1) (blocks indefinitely)
yes
EINTR?
no
fds[0] wakeup?
POLLIN
no event
drain_wakeup() (read all bytes, discard)
run_posted(): swap posted_ under lock execute all lambdas
running_?
false
true
return from run()
fds[1] new conn?
POLLIN
no event
accept_client(): accept(), O_NONBLOCK TCP: TCP_NODELAY + SO_KEEPALIVE clients_[fd] = {}
fds[2..N] client data?
POLLIN/HUP/ERR
handle_client_data(fd)
handle_client_data(fd)
read(fd, buf, 4096)
n <= 0
n > 0
remove_client(fd): close(fd) clients_.erase(fd)
Append to client read_buf
Contains newline?
yes
no
Extract line
done with this fd
parse_ipc_message(line)
not Request
Request
send InvalidRequest error
Look up handlers_[method]
not found
found
send MethodNotFound error
Invoke handler(req, resp, err)
true
false
send_to(fd, serialize(resp))
send_to(fd, serialize(err))

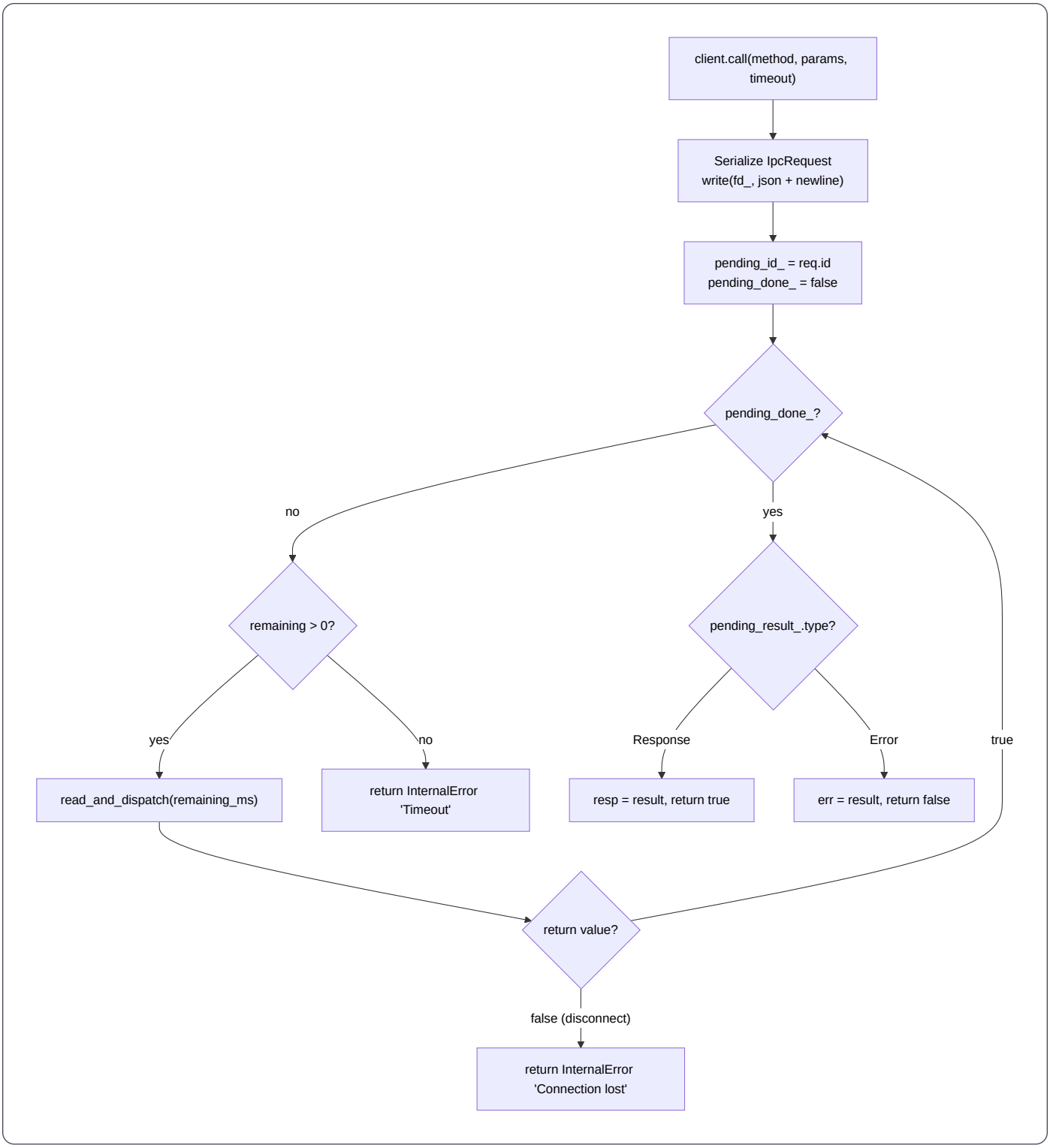
5. IPC Client Flow

5a. Connection

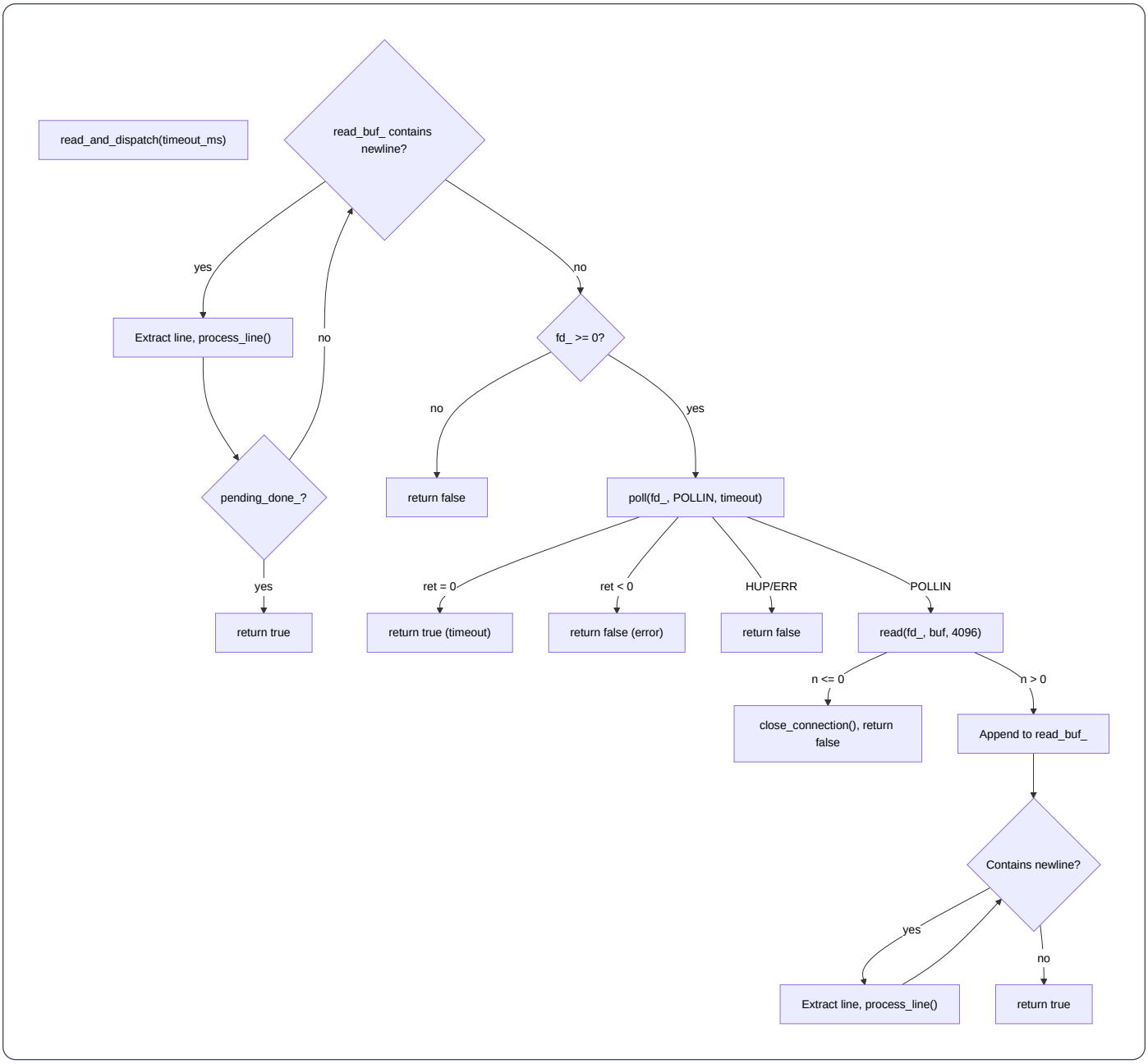

client.connect()
fd_ >= 0?
yes
no
return true
(already connected)
addr_.transport?
TCP
socket(AF_INET)
set O_NONBLOCK
connect() ->
EINPROGRESS
Unix
poll(POLLOUT, 5000 ms)
ready
timeout
getsockopt(SO_ERROR)
0
error
Restore blocking mode
TCP_NODELAY
SO_KEEPALIVE 30s/10s/3
close, return false
socket(AF_UNIX)
connect(sock_addr)



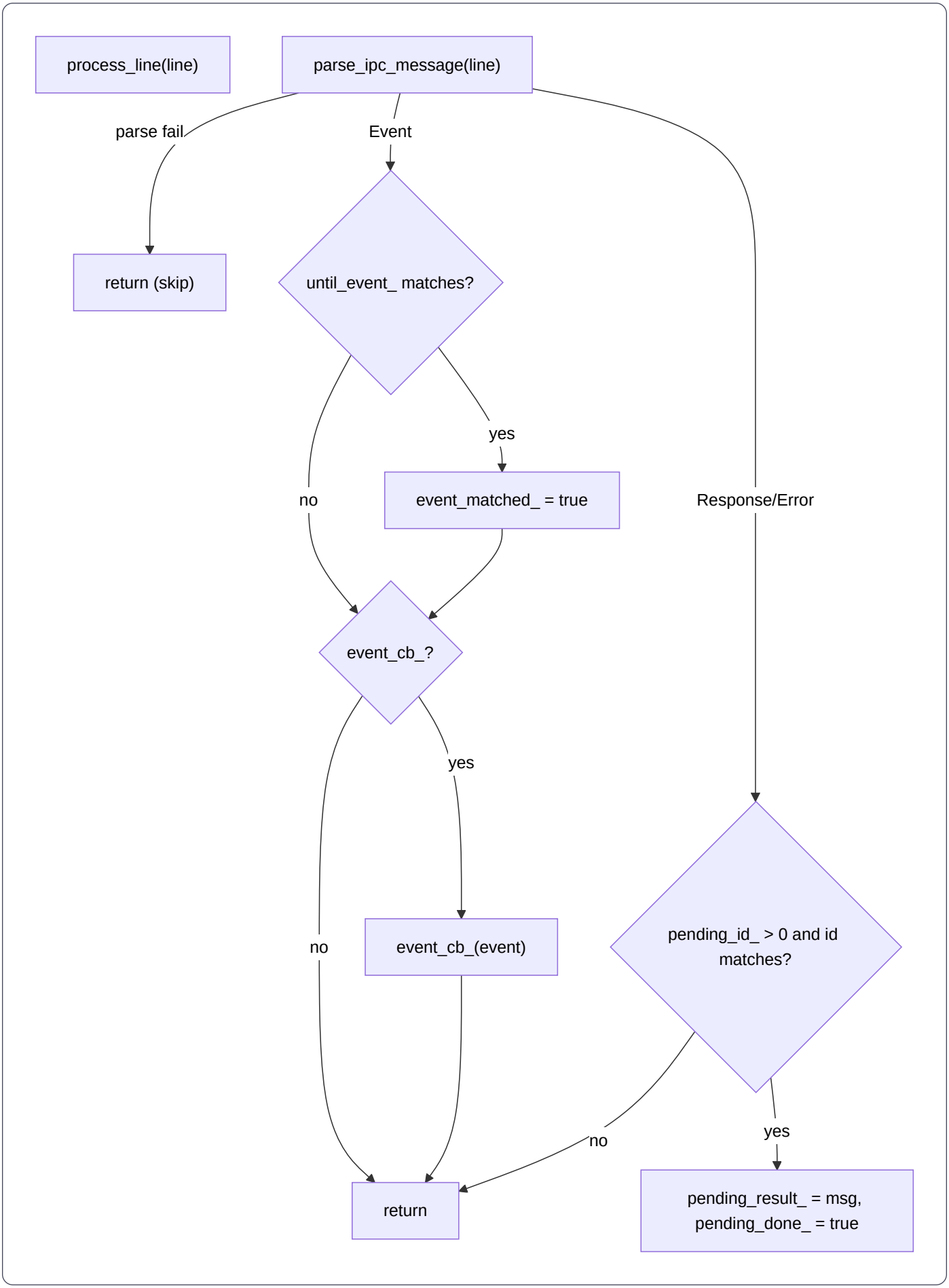
5b. Blocking RPC (call)



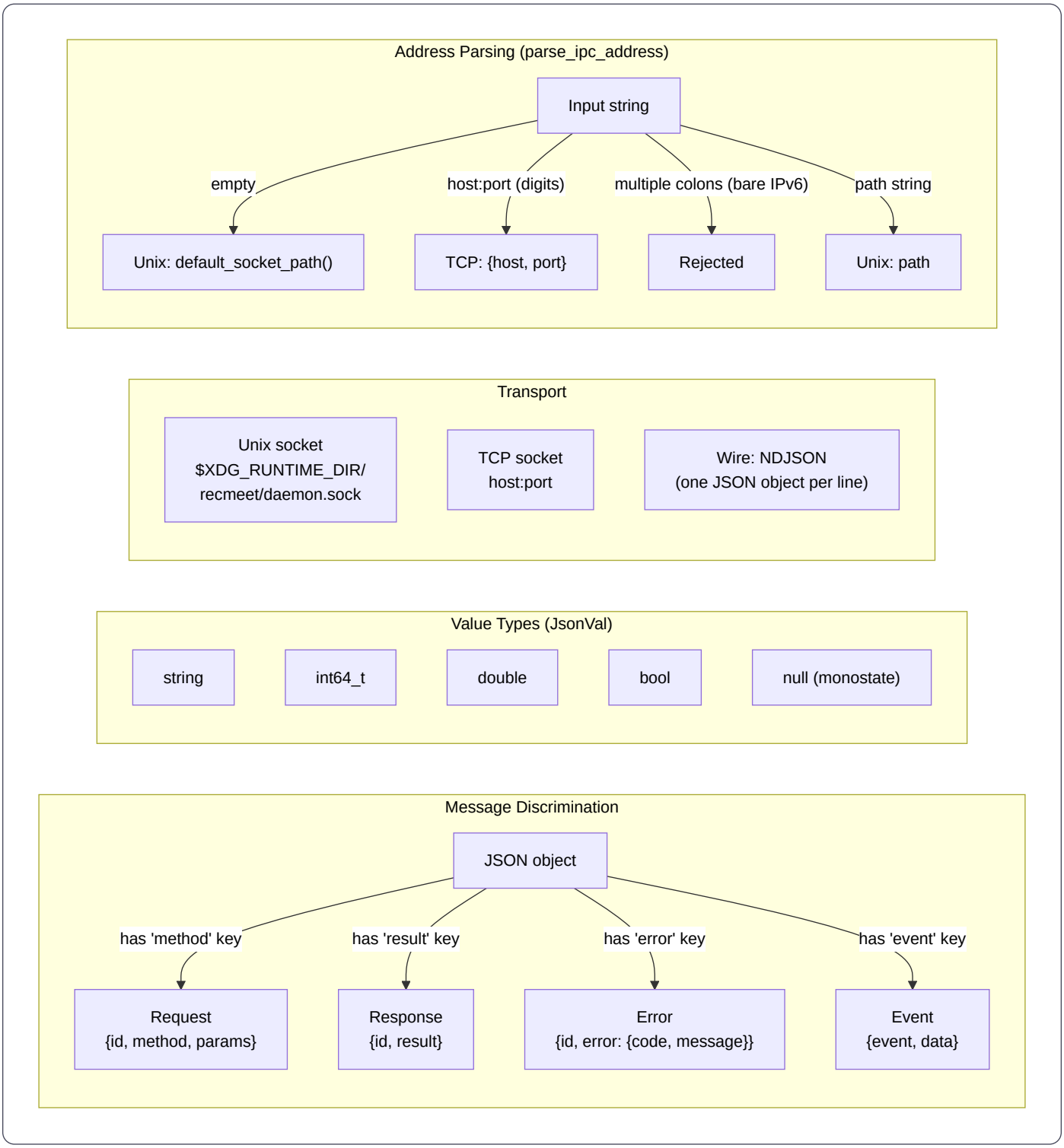
5c. read_and_dispatch() Internal



5d. process_line() Dispatch

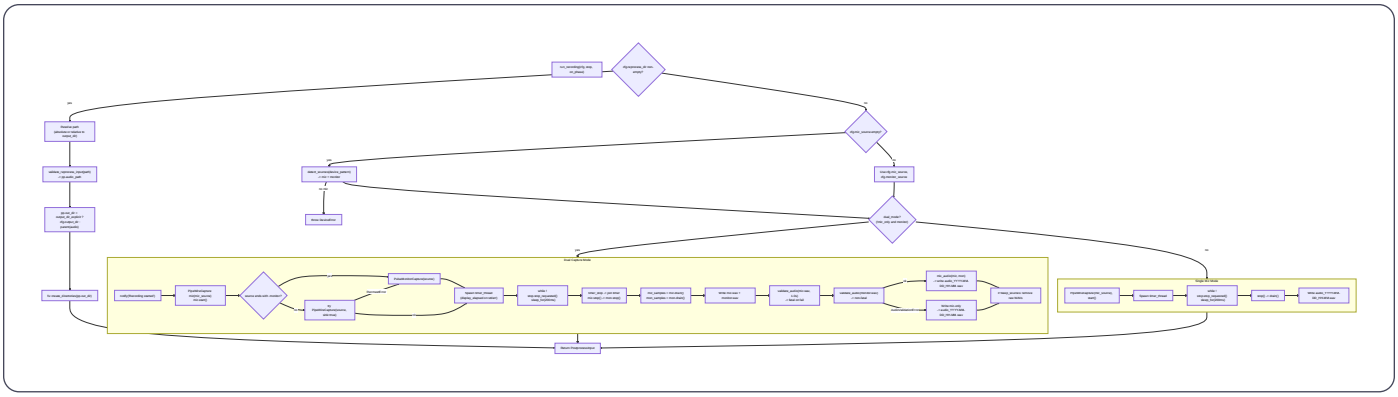

process_line(line)
parse_ipc_message(line)
parse fail
Event
return (skip)
until_event_ matches?
yes
no
event_matched_ = true
Response/Error
event_cb_?
yes
no
event_cb_(event)
pending_id_ > 0 and id matches?
no
yes
return
pending_result_ = msg, pending_done_ = true

6. IPC Protocol Wire Format



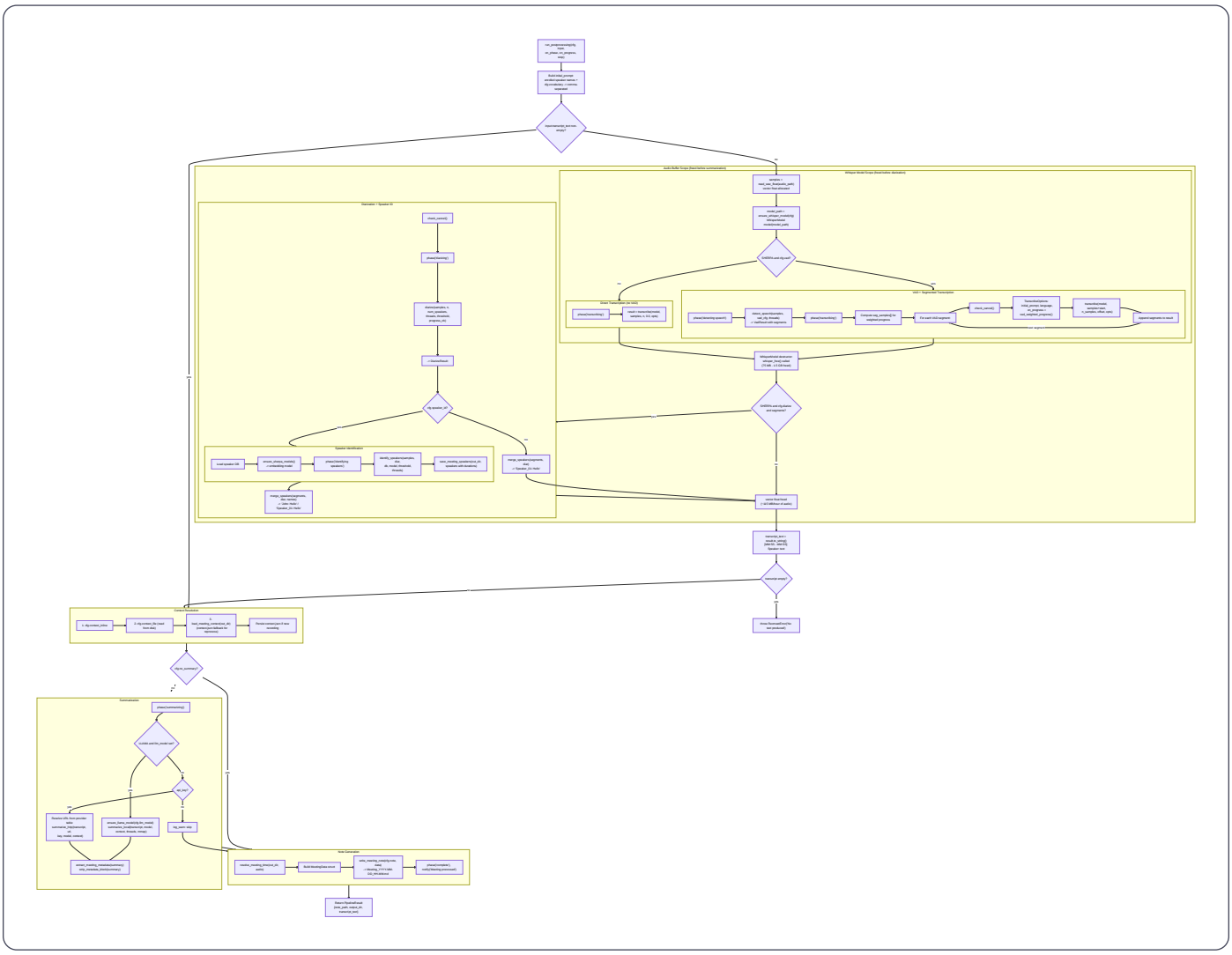
7. Recording Pipeline

`run_recording()` control flow with dual-capture and reprocess branching.



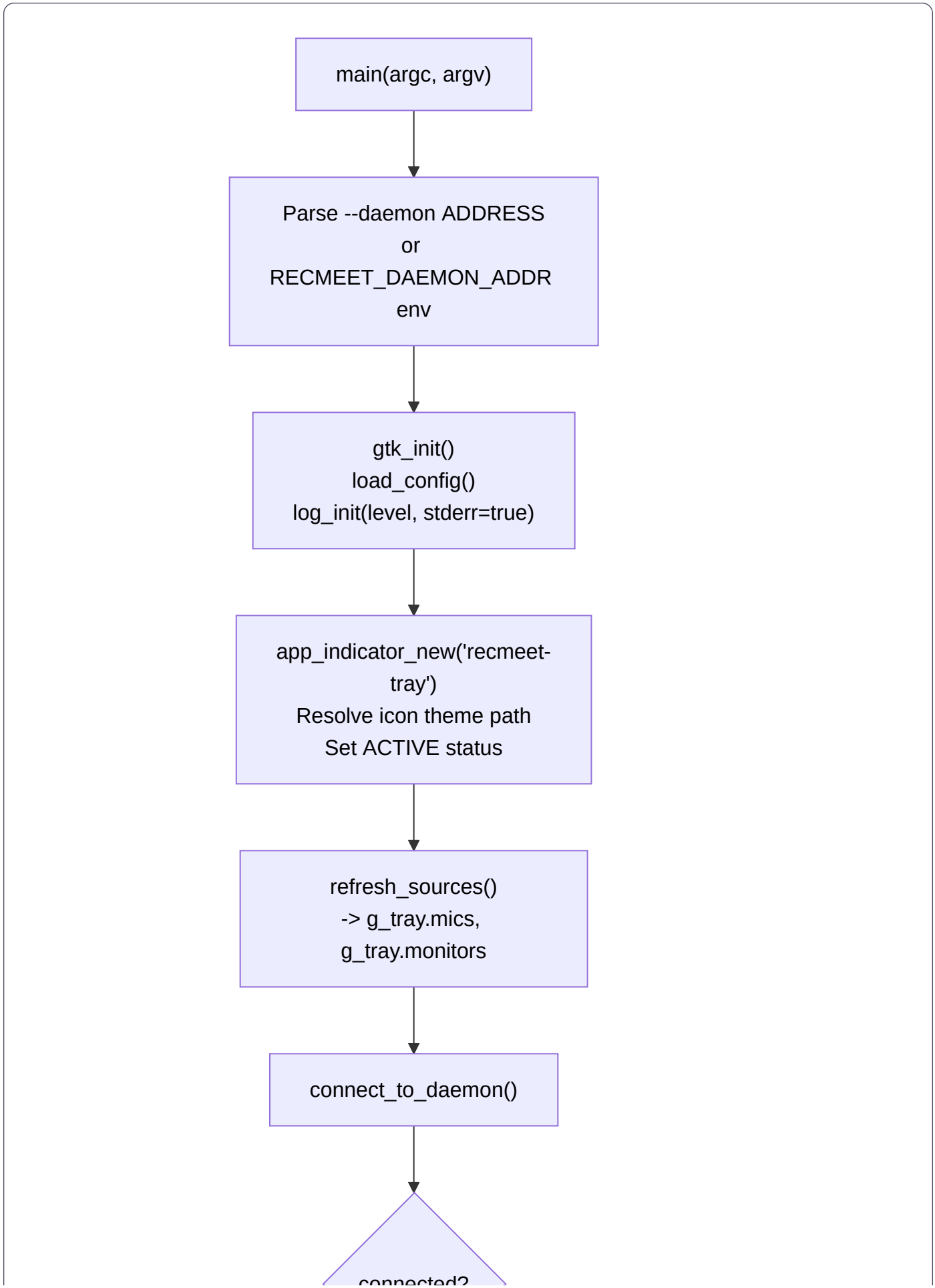
8. Postprocessing Pipeline

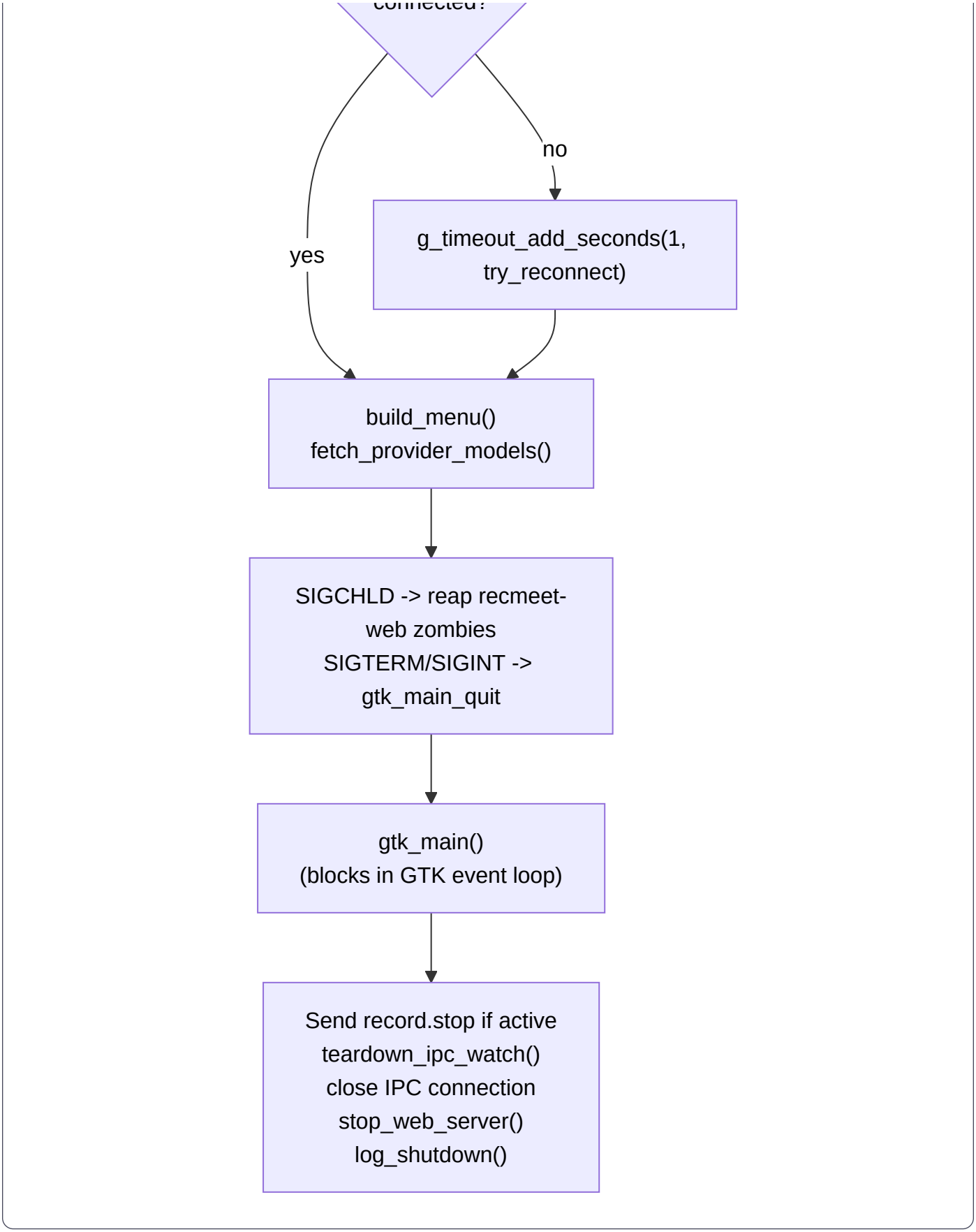
`run_postprocessing()` with memory scoping, cancellation, and all ML stages.



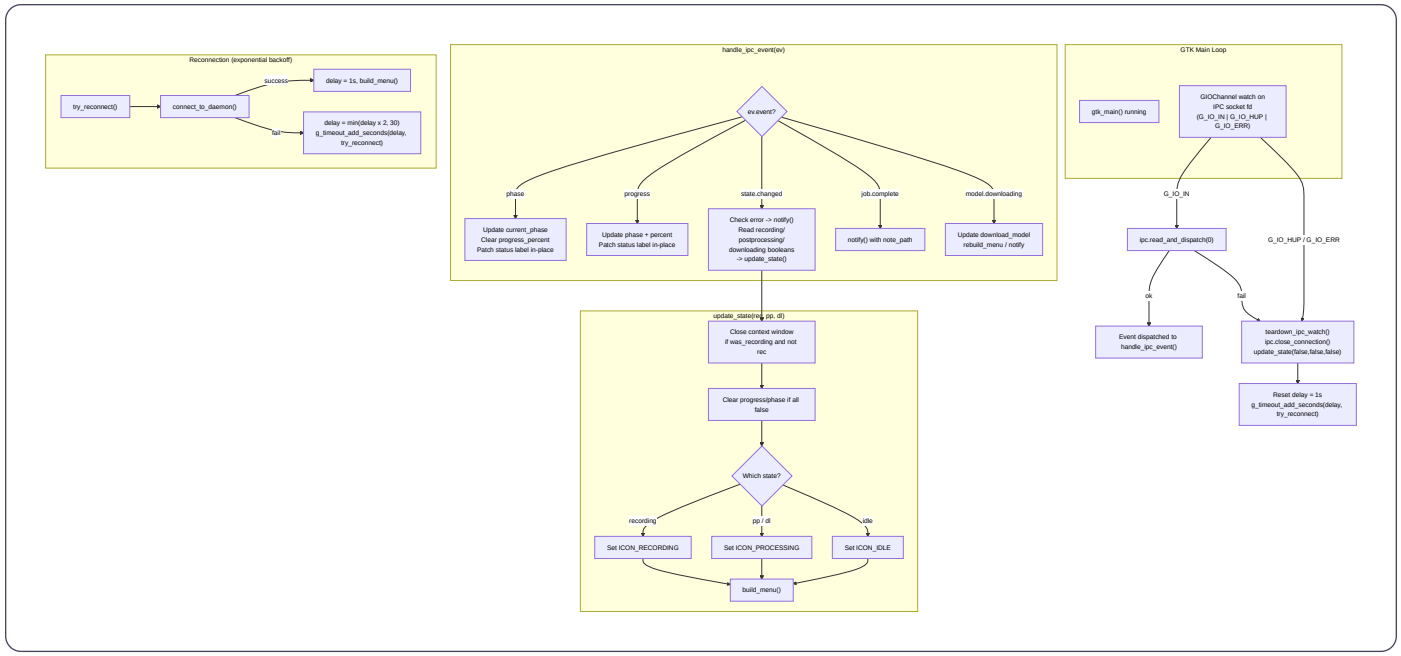
9. Tray Applet

9a. Tray Startup and GTK Main Loop

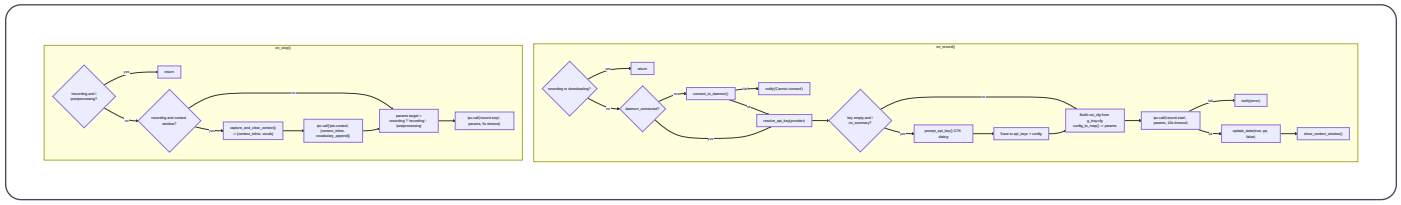

main(argc, argv)
Parse --daemon ADDRESS or RECMEET_DAEMON_ADDR env
gtk_init()
load_config()
log_init(level, stderr=true)
app_indicator_new('recmeet-tray')
Resolve icon theme path
Set ACTIVE status
refresh_sources()
-> g_tray.mics, g_tray.monitors
connect_to_daemon()
connected?



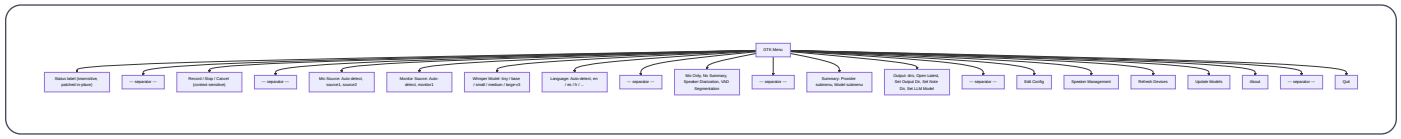
9b. IPC Event Integration with GTK



9c. Recording Start/Stop Flow



9d. Tray Menu Structure



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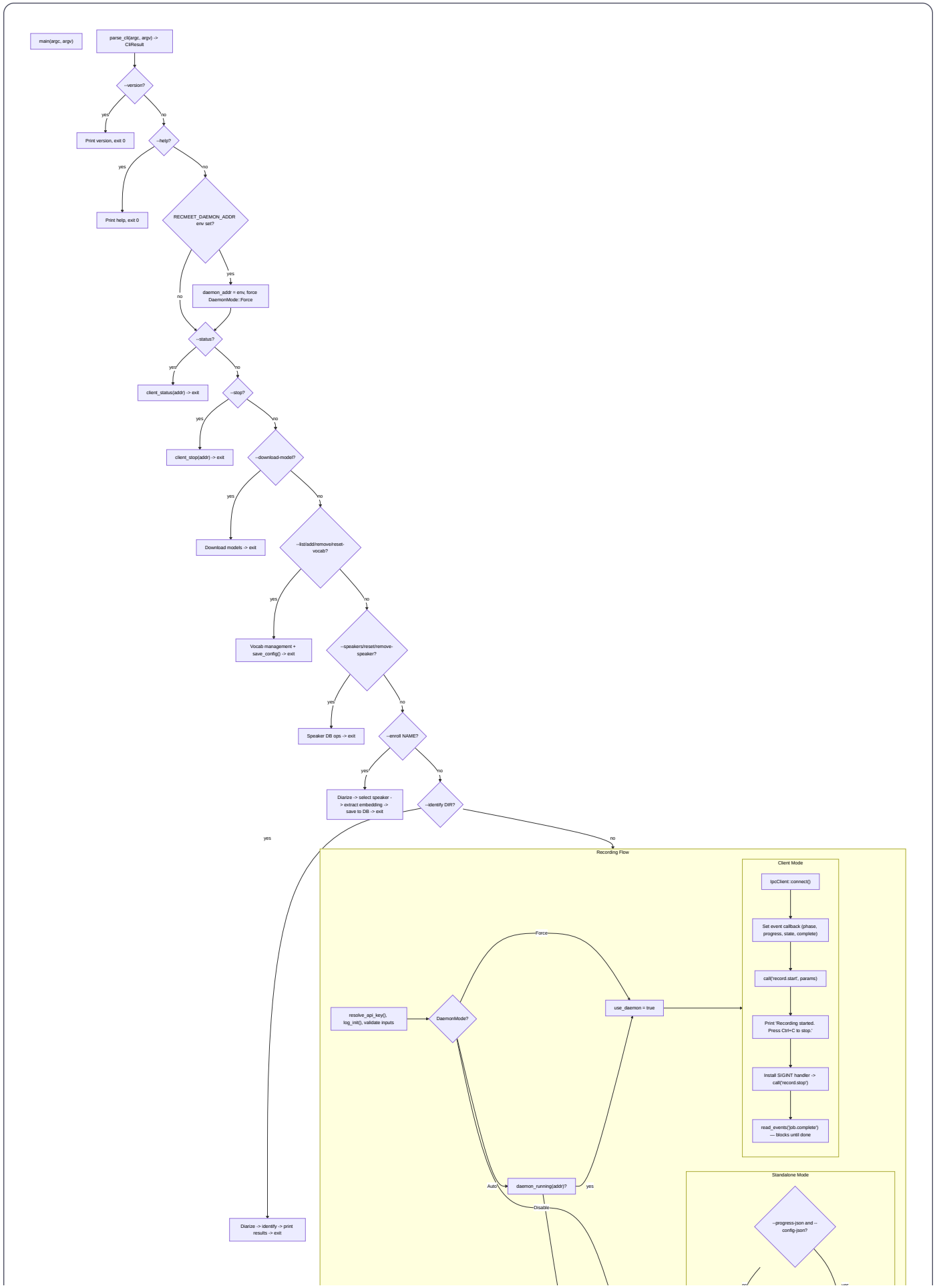
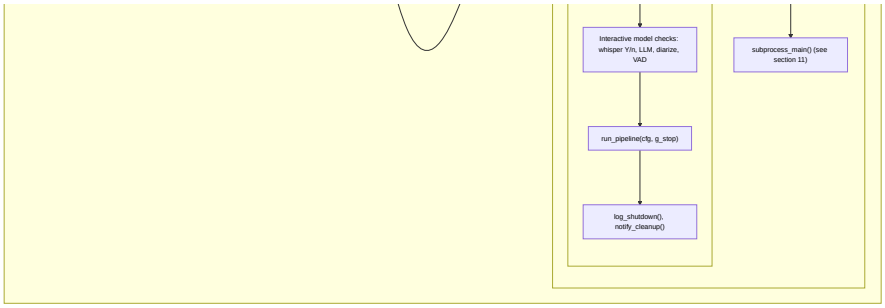
10. CLI Mode Selection

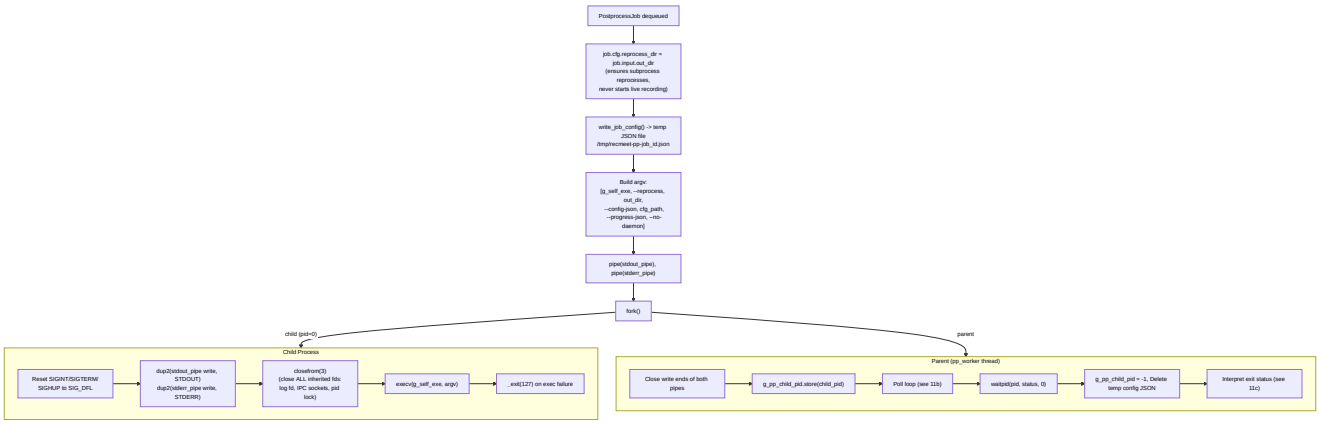

main(argc, argv)
parse_cli(argc, argv) -> CliResult
--version?
yes
no
Print version, exit 0
--help?
yes
no
Print help, exit 0
RECMEET_DAEMON_ADDR env set?
yes
no
daemon_addr = env, force DaemonMode::Force
--status?
yes
no
client_status(addr) -> exit
--stop?
yes
no
client_stop(addr) -> exit
--download-model?
yes
no
Download models -> exit
--list/add/remove/reset-vocab?
yes
no
Vocab management + save_config() -> exit
--speakers/reset/remove-speaker?
yes
no
Speaker DB ops -> exit
--enroll NAME?
yes
no
Diarize -> select speaker -> extract embedding -> save to DB -> exit
--identify DIR?
yes
no
Diarize -> identify -> print results -> exit
Recording Flow
resolve_api_key(), log_init(), validate inputs
DaemonMode?
Force
Auto
Disable
daemon_running(addr)?
yes
use_daemon = true
Client Mode
IpcClient::connect()
Set event callback (phase, progress, state, complete)
call("record.start", params)
Print 'Recording started. Press Ctrl+C to stop.'
Install SIGINT handler -> call("record.stop")
read_events("job.complete") — blocks until done
Standalone Mode
--progress-json and --config-json?



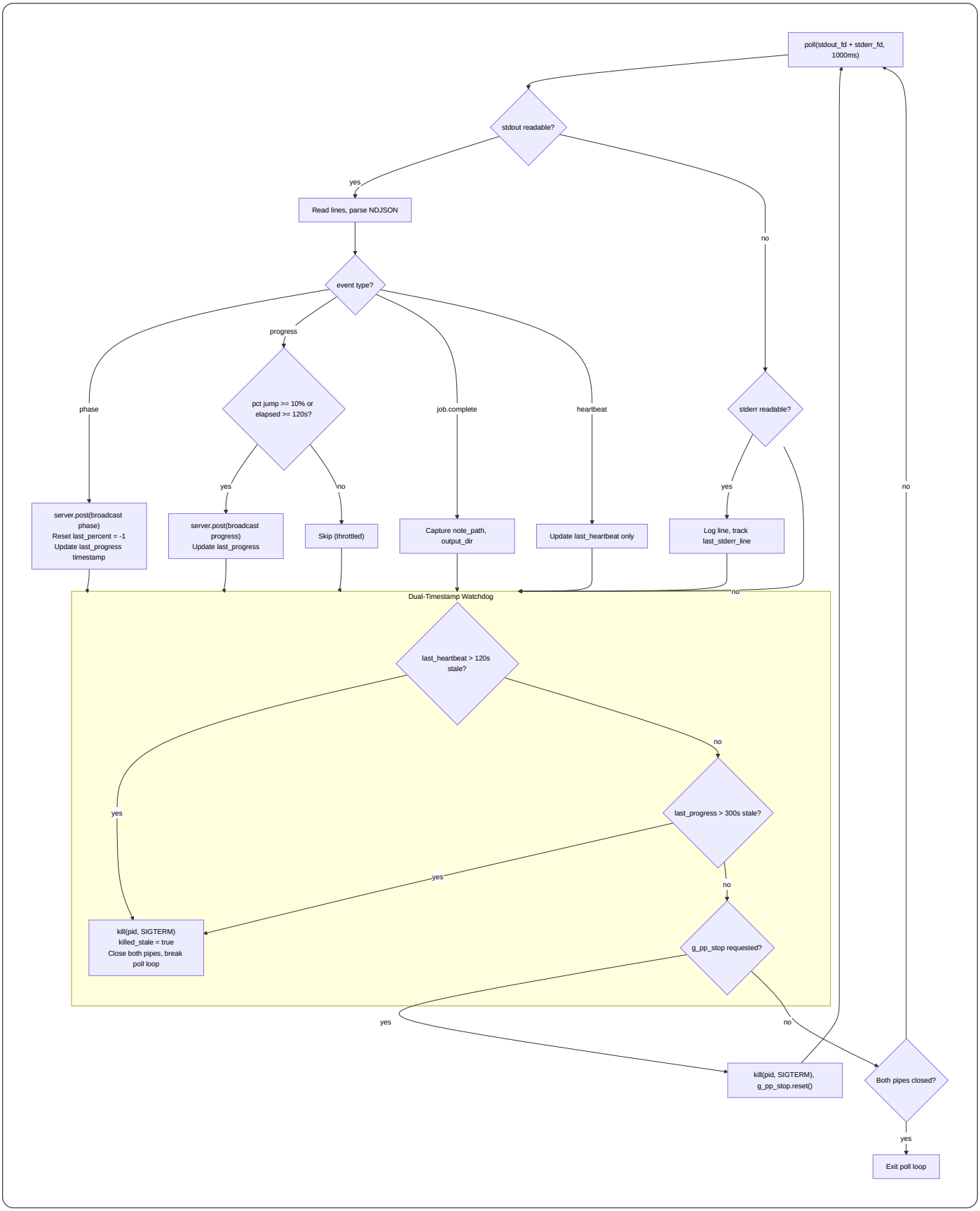
11. Subprocess Postprocessing

The daemon's `pp_worker_loop` fork/exec's the `recmeet` binary as a child process for crash isolation. Communication is via NDJSON on stdout/stderr pipes.

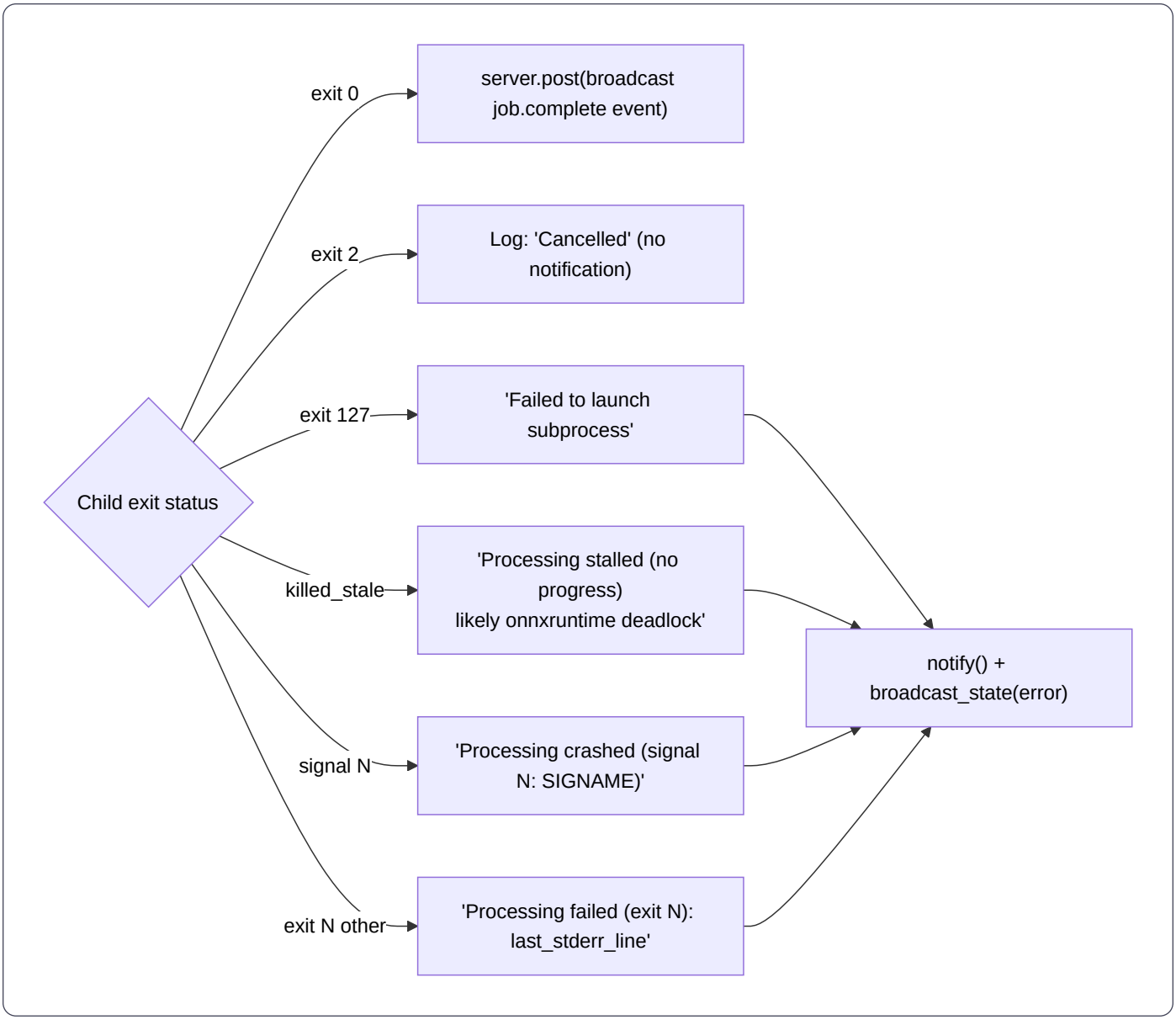
11a. Fork/Exec Flow



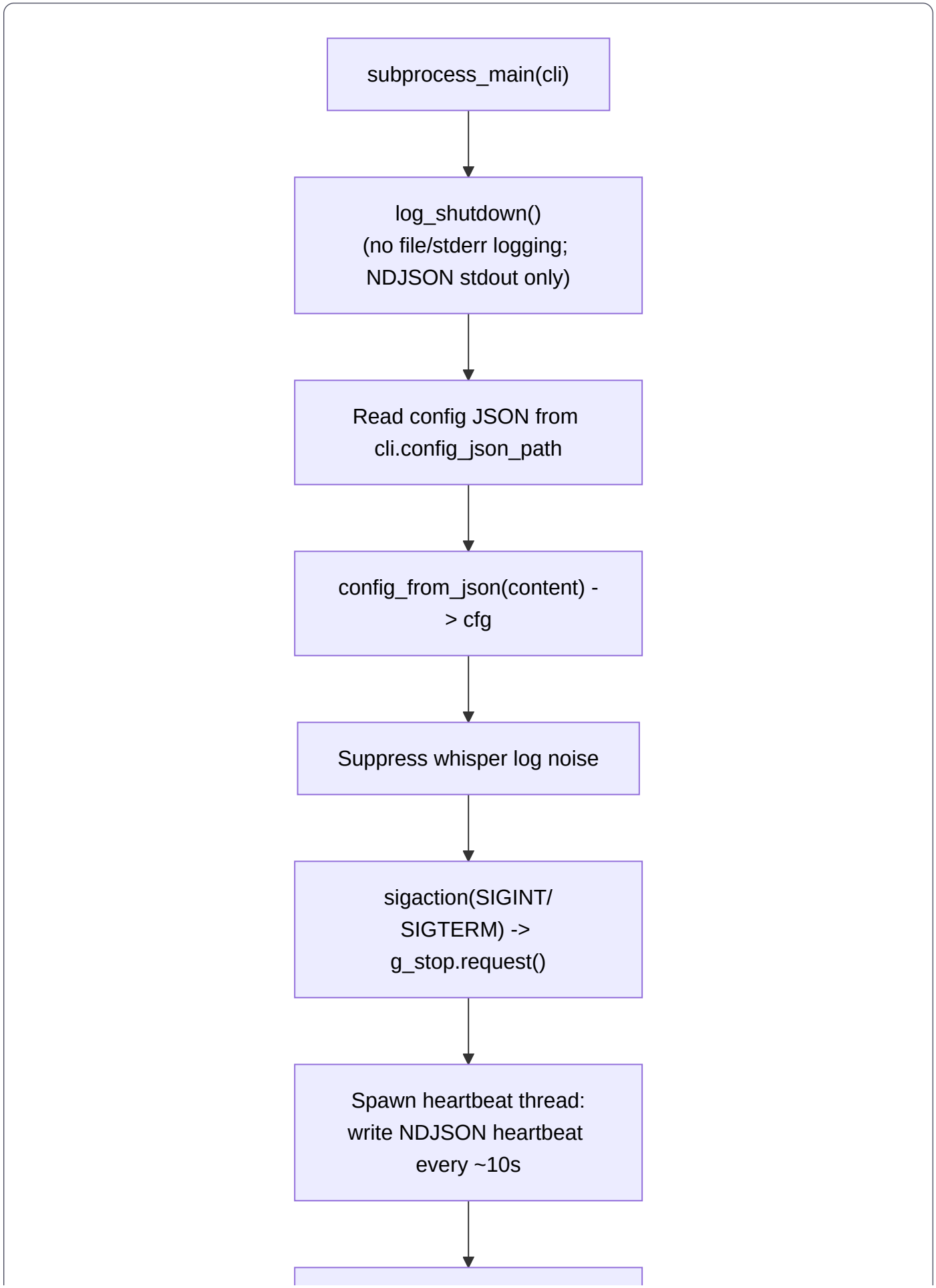
11b. NDJSON Poll Loop and Watchdog

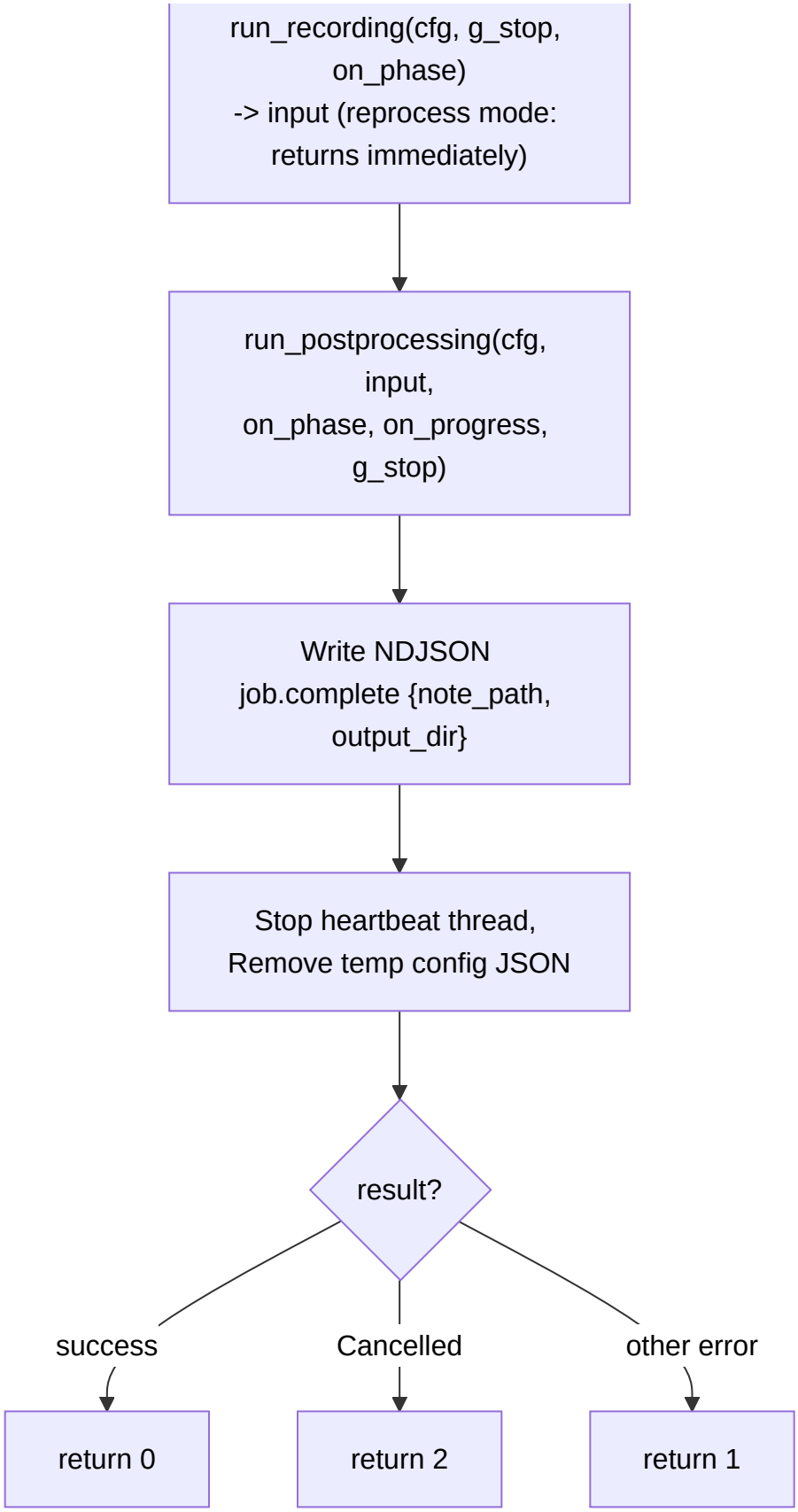


11c. Exit Status Interpretation

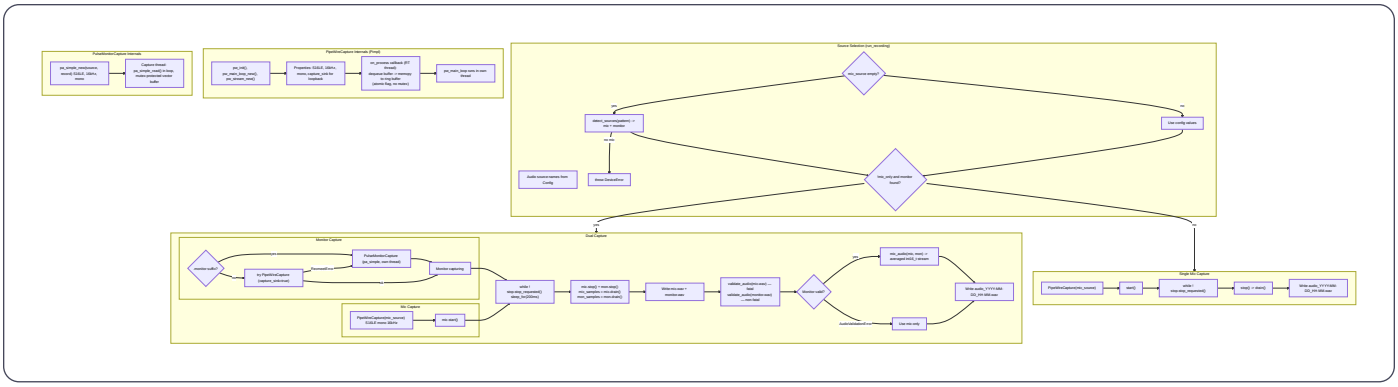


11d. Subprocess Internal Flow


subprocess_main(cli)
log_shutdown()
(no file/stderr logging;
NDJSON stdout only)
Read config JSON from
cli.config_json_path
config_from_json(content) -
> cfg
Suppress whisper log noise
sigaction(SIGINT/
SIGTERM) ->
g_stop.request()
Spawn heartbeat thread:
write NDJSON heartbeat
every ~10s



12. Audio Capture Subsystem



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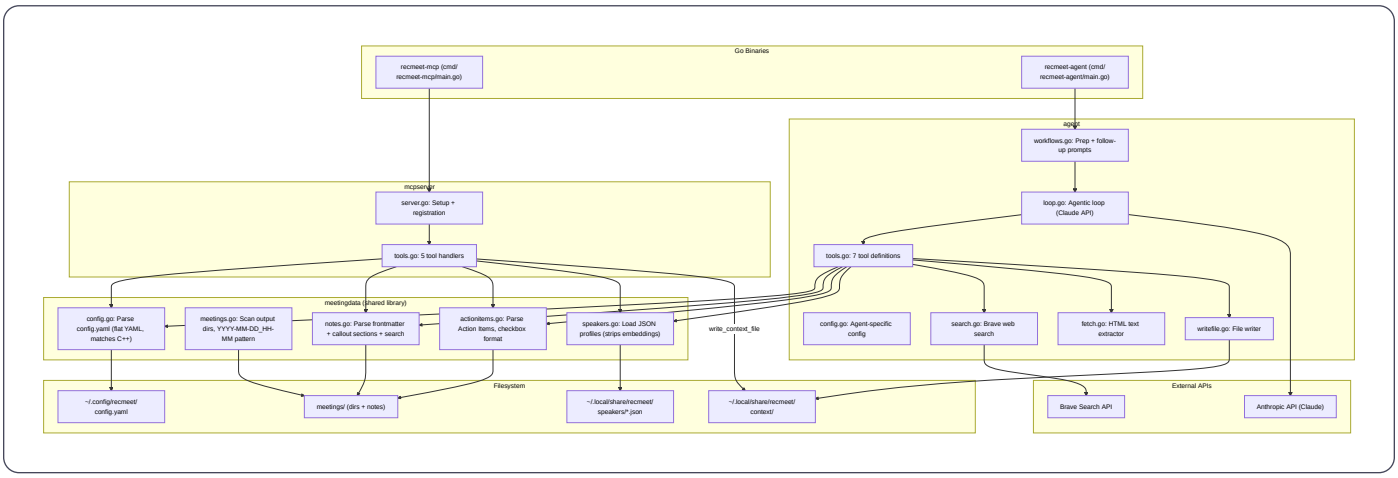
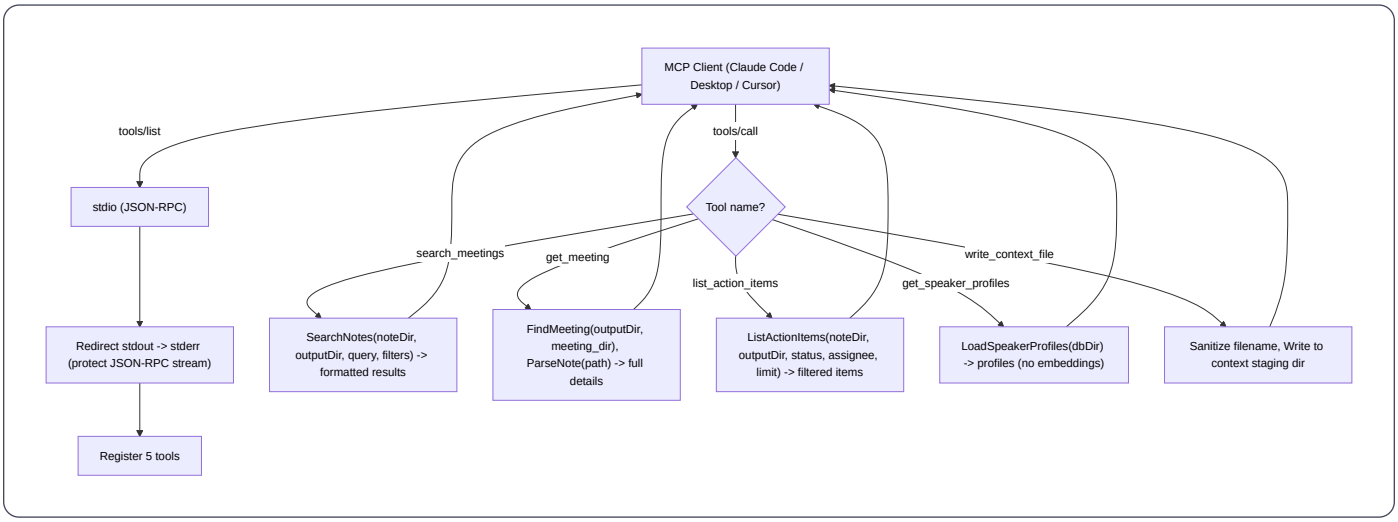
13. Go Tools Module

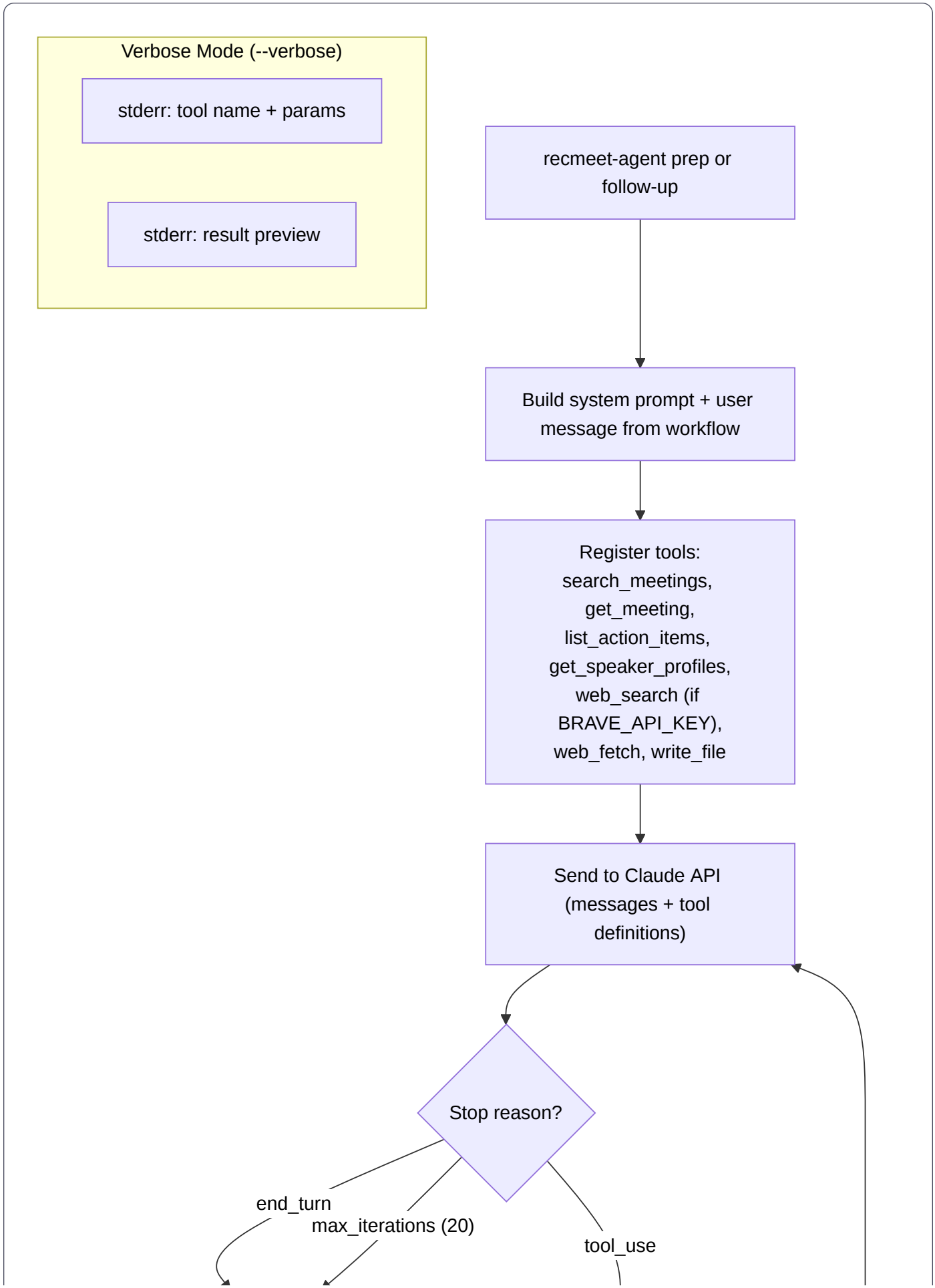
13a. Module Structure and Data Flow

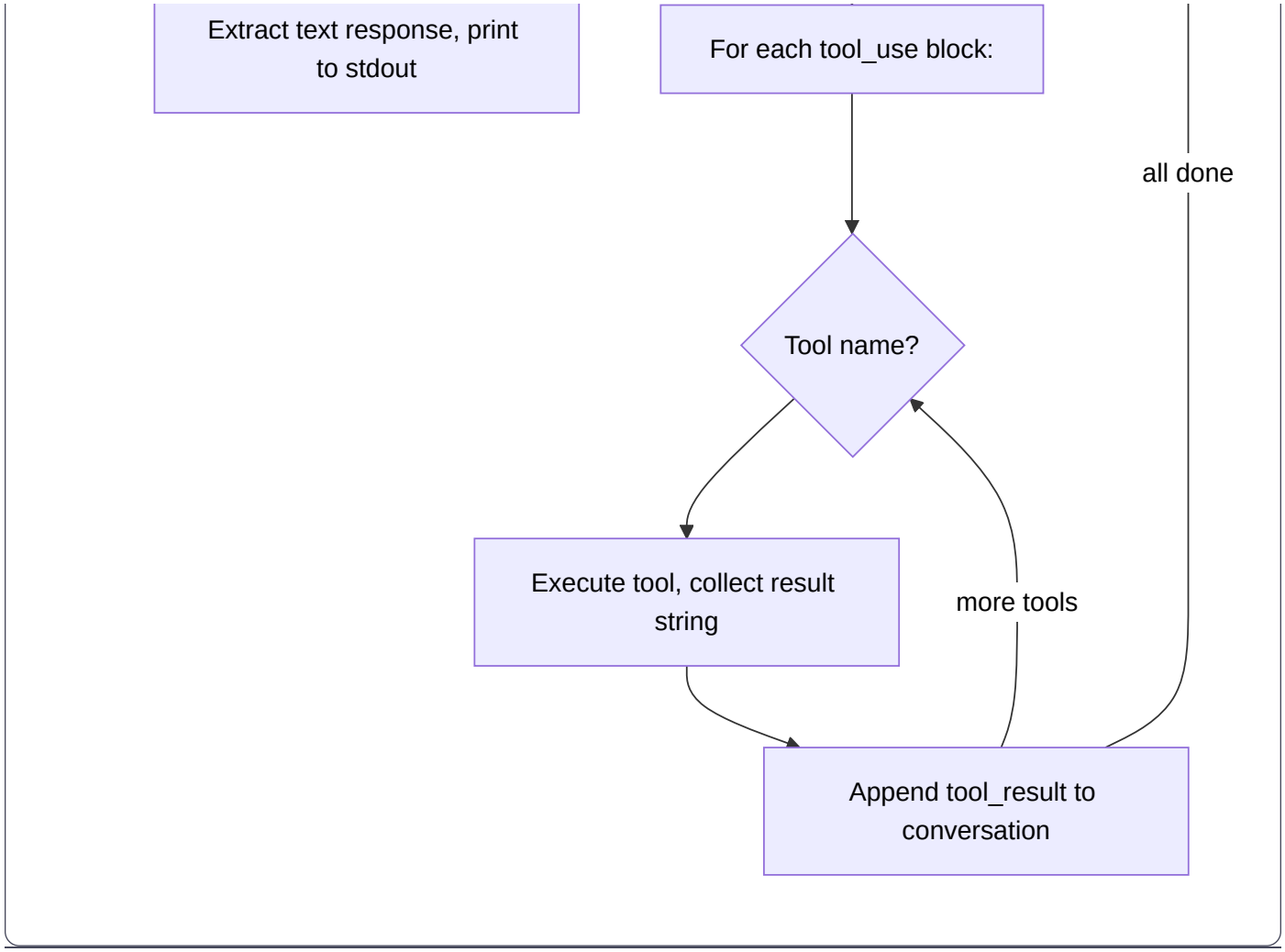


13b. MCP Server Tool Dispatch



13c. Agent Agentic Loop


Verbose Mode (--verbose)
stderr: tool name + params
stderr: result preview
recmeet-agent prep or follow-up
Build system prompt + user message from workflow
Register tools: search_meetings, get_meeting, list_action_items, get_speaker_profiles, web_search (if BRAVE_API_KEY), web_fetch, write_file
Send to Claude API (messages + tool definitions)
Stop reason?
end_turn
max_iterations (20)
tool_use



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